

Video2Song: Video-to-Song Generation via Retrieval-Augmented Prompt Generation

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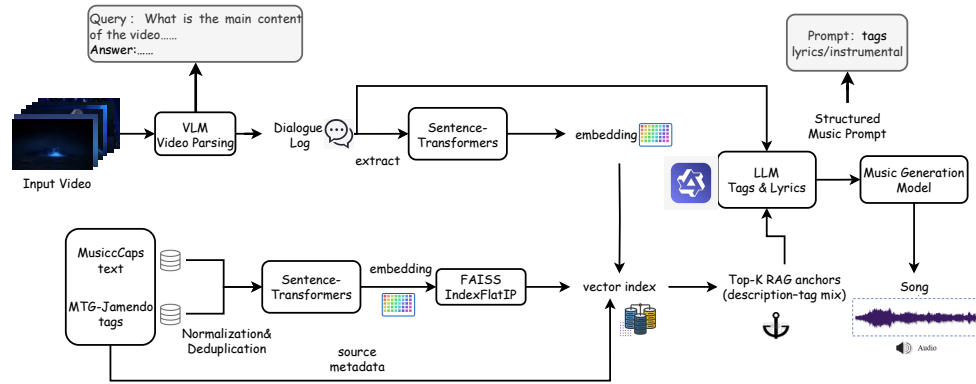


Figure 1: We present an end-to-end framework, Video2Song, for video soundtrack generation that combines a VLM and retrieval-augmented generation to extract semantic, stylistic, and rhythmic cues, conditioning a LLM to guide music synthesis.

Abstract

In this paper, we introduce an end-to-end framework for generating video soundtracks that enhances content alignment and controllability compared to existing approaches. Leveraging a video-language model (VLM) to interpret video semantics and retrieval-augmented generation (RAG) to extract compatible style and tempo cues from a music corpus, our framework conditions a large language model (LLM) to guide a music generator in producing high-fidelity soundtracks. Evaluations across diverse video genres demonstrate superior semantic alignment and rhythmic synchronization, as well as the framework’s effectiveness and generalizability. Project page: <https://github.com/zijiao1108/Video2Song>.

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1 Introduction

Video soundtrack generation plays a crucial role in enhancing narrative immersion, emotional impact, and perceptual coherence in visual media experiences. In recent years, music generation models have advanced significantly, achieving high-quality and high-fidelity synthesis [Li et al. 2025]. Nevertheless, within the specific context of video soundtrack generation, these models often exhibit limited controllability. Without explicit guidance, they may yield outputs that diverge from the video content, a phenomenon commonly referred to as “hallucination”, which substantially degrades contextual alignment. To address this issue, we introduce a novel end-to-end framework that integrates the video understanding capabilities of a video language model (VLM) [Zhang et al. 2025] with the external knowledge retrieval provided by retrieval-augmented generation (RAG) [Lewis et al. 2020], thereby reducing the opacity of purely generative methods. This mechanism grounds the music generation process in a human-annotated corpus enriched with descriptive narratives and tag-based metadata, enhancing controllability and diminishing the large language model (LLM)’s tendency to hallucinate in fine-grained stylistic dimensions.

2 Method

Our proposed framework is illustrated in Figure 1. The process begins with a VLM (TinyLLaVA-Video [Zhang et al. 2025]), which performs a templated and controllable multi-turn question–answering procedure on the input video. This interaction produces a dialogue log that captures plot complexity, narrative structure, and emotional tone. Rather than functioning solely as a summary, the dialogue log serves as a central textual resource for downstream lyric generation and content selection, while simultaneously providing semantic and rhythmic cues that support retrieval and conditional control.

We then adopt a Retrieval-Augmented Generation (RAG) strategy, building a searchable vector library from two complementary, human-annotated resources. The first contains the free-text music descriptions from MusicCaps [Agostinelli et al. 2023], while the second contains genre/instrument/emotion tag combinations from MTG-Jamendo [Bogdanov et al. 2019]. We retain the original descriptions from MusicCaps and standardize, de-duplicate, and merge the Jamendo music tags into a unified tag string. Each entry is then encoded as an L2-normalized vector using Sentence-Transformers (all-mpnet-base-v2) [Reimers and Gurevych 2019] and indexed with FAISS IndexFlatIP to enable efficient inner-product retrieval, with source metadata retained for contextual reference. The resulting “description–tag” hybrid evidence base encompasses both narrative and emotional semantics, as well as production-level attributes such as genre, instrumentation, and tempo.

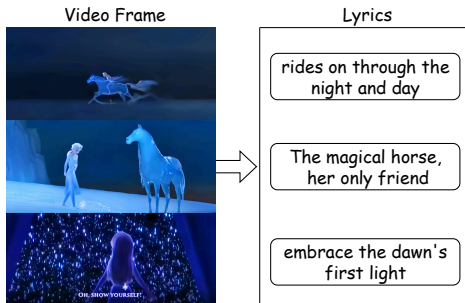


Figure 2: An example of video–Lyric alignment result.

For retrieval, the system transforms the dialogue log into phrase-level queries by extracting and normalizing key information. To reconcile free-form textual descriptions with the structured tag schema, we augment the queries with genre, instrument, tempo, and emotion keywords aligned to the schema. In addition, the system jointly infers the vocal strategy (lyric-centric or purely instrumental) and tempo preference, where available BPM values are coarsely mapped to slow, medium, or fast categories. The resulting query vectors are encoded with the same model and matched within the index. Candidate results are re-ranked using a lightweight procedure: beyond fundamental similarity, the ranking favors items consistent with the inferred vocal and tempo preferences, while appropriately weighting matches for prominent instruments and emotional terms. Conversely, candidates with excessively long tags or conflicting signals are penalized. To mitigate corpus-specific bias, results from MusicCaps and Jamendo are interleaved, and the top- K entries are retained as the “RAG evidence anchors”.

Subsequently, we invoke an LLM to synthesize a structured music prompt. In experiments, the instruction template supplied to the LLM is iteratively refined, with retrieved evidence anchors, covering attributes such as musical style, tempo, and primary instrumentation, embedded into designated slots within the template. By fixing the primary music tag information, the system mitigates stylistic drift and rhythmic instability that may occur in free-form prompt generation. The decision of whether to generate lyrics, along with the thematic framing and lyrical cues, is automatically inferred by the LLM from the video dialogue log, thereby ensuring consistency with the underlying video narrative. Finally, the generated prompt is passed to a music generation model, ACE-Step [Gong et al. 2025], to synthesize the final soundtrack.

3 Results and Conclusion

Figure 2 presents qualitative examples illustrating the semantic consistency between selected video frames and the automatically generated lyrics, thereby providing an intuitive demonstration of semantic alignment. We conducted evaluations across diverse video categories, including landscapes, music videos, feature films, and animations (see supplementary video). With all other settings held constant, the RAG module substantially improved the quality of LLM-generated music prompts, yielding closer alignment with video narrative semantics and affective tone. This integration also reduced irrelevant elements and mitigated stylistic drift. Moreover, modifying or expanding the RAG corpus enabled predictable modulation of the generated musical style and content, thereby demonstrating enhanced controllability. In the current work, a lightweight VLM was employed for video understanding. Our findings suggest that adopting a more powerful VLM could further refine the extraction of video semantics and rhythmic cues, thereby providing higher-quality queries for the RAG module. In conclusion, we present a novel framework that integrates a retrieval-augmented generation module with LLMs, thereby enhancing semantic alignment, affective consistency, and controllability in video-to-song generation. Future work will focus on leveraging stronger VLMs and detailed quantitative metrics to further improve retrieval quality and systematically evaluate performance.

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